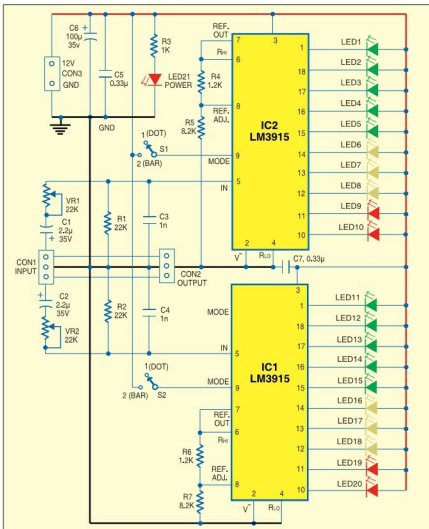




fier. CON1 is the input connector and CON2 output connector for the loudspeaker or headphone. Each channel

has its own LM3915 and ten LEDs to indicate the power level. To indicate the different audio levels, select LEDs

of three colours as per your liking. For example, you can have five green LEDs, three yellow LEDs and two red LEDs.

If appropriate signal generators and measuring equipment are not available, the level indicator can be calibrated based on personal observations. For example, the audio signal should be in the green LEDs zone when the signal is strong enough but not irritating, in the yellow zone when it is disturbing or starting to get distorted, and in the red zone when it is heavily distorted or too strong to listen to in the room.

Calibration can be done with the help of potentiometers VR1 and VR2, which are optional. Switches S1 and S2 let you select between two modes of LM3915 operation—the bar mode and the dot mode. These too can be removed or replaced with jumpers, if not required. When the switches are removed, leave pins 9 of IC1 and IC2 open.

The circuit works off regulated 12V. You can also power it through a 12V rechargeable battery.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix all the LEDs in two rows on the front side of the cabinet. Also affix the two terminals for input and output on rear side of the cabinet. ●

FM Bug

S.C. DWIVEDI

This FM bug transmitter circuit will let you spy on people. The transmitter can be placed in the desired room and the conversation heard from a place far away just using a regular FM radio set.

The circuit is designed around a single transistor 2N3904 (T1), a custom-made coil (L1), three capacitors (C1 through C3), a trimmer (VC1), two

resistors (R1 and R2) and, of course, a condenser microphone (MIC1). The circuit transmits in the frequency range of 88-105 MHz. Transmission range is 100 metres.

Working of the circuit is simple and based on analogue modulation in which a carrier signal is varied corresponding to the message signal.

The microphone picks up the sounds in its vicinity to produce corresponding electrical signal. This is



the message signal that needs to be transmitted over FM band. The message signal is fed to the base of transistor T1. The tank circuit made using trimmer VC1 and coil L1 generates the carrier frequency. This frequency can be tuned using the trimmer. Transistor T1 modulates the audio signal from condenser microphone over the carrier signal produced by tank circuit. This